

**Department of Software Engineering
(Formal Methods)**

Assignment 02

Semester: Spring 2026 (FM)

Course Instructor: Ms. Nigar Azhar Butt

Sections: All

Instructions

- Read the instructions very carefully.
- Late assignments will not be accepted
- **Plagiarism may result in zero marks in the whole assignments category (all assignments) regardless of the percentage plagiarized.**
- There will be no credit if the given requirements are changed.
- This is an individual assignment, meaning you must complete it on your own.

Global Submission Instructions (Applies to Q1 – Q4)

- **Handwritten Requirements (Scanned PDF Submission):**
 - **FOL Translations:** For questions (Question 2 through Question 4), the identification of Constants, Variables, Predicates, and Functions, as well as the FOL formulas, must be **handwritten**.
 - **Solver Analysis:** For all question, you must state the solver's result (sat/unsat). If **sat**, you must mention and explain the resulting values. If **unsat**, provide a handwritten explanation of which logic/constraints caused the contradiction.
 - **Logic Code:** You must provide handwritten code for **any two parts** from **Question 1 and Question 2**.
 - **File Format:** All handwritten work must be scanned and submitted as a single PDF named <Rollnumber_Section>_solution.pdf.

- **Digital File Submissions (.smt2):**
 - For all parts of Question 1, Question 2, Question 3, Question 4, and Question 5, submission must be in original .smt2 files; all parts are needed.
 - **Naming:** Use the convention <Rollnumber_Section>_Qx_part.smt2.
- **Screenshot PDF:**
 - A single combined PDF containing screenshots of Z3 solver outputs for all questions is required.
 - Your **Roll Number and Name** must be clearly visible in every screenshot.
 - **Naming:** <Rollnumber_Section>_screenshots.pdf.

Question 1:

For the following, provide code in SMT-LIB syntax and screenshots of corresponding solutions proving satisfiability and validity.

1. $(\neg p \wedge (q \rightarrow r)) \rightarrow ((\neg s \vee t) \wedge (q \rightarrow p))$
2. $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$
3. $(q \rightarrow r) \wedge (r \rightarrow s) \wedge (s \rightarrow t) \wedge (t \rightarrow \neg q) \wedge q$
4. $((p \wedge q) \vee (r \wedge s)) \rightarrow ((p \vee r) \wedge (q \vee s))$

Question 2:

For each statement you are required to perform the following task using First Order Logic (FOL).

- A) Identify Constants.
- B) Identify Variables.
- C) Identify Predicates.
- D) Identify Functions.

You are required to translate each statement into First Order Logic (FOL) formulas with the correct respective quantifiers. And provide Code as instructed.

1. Code Submission & Screenshots:

- Provide SMT-LIB code files (Z3) that models the logic and proves the goal.
- Submit screenshots of the SMT-solver output confirming the satisfiability and the specific result.

Scenario 1: The Production Server Crash

There are three entities in the server room: the **Admin**, the **Intern**, and the **Automated Bot**. One of them triggered a `rm -rf /` command that crashed the production server.

- Someone who triggered the command was logged in at 3:00 AM.

- The only entities logged in at 3:00 AM were the **Intern** and the **Admin**.
- An entity is "reckless" if they trigger commands without a peer review.
- The **Admin** is never reckless.
- The **Automated Bot** only triggers commands if they are peer-reviewed.
- The **Intern** triggers a command if and only if the **Admin** is not logged in.
- **Query:** Who crashed the server?

Scenario 2: The Data Breach Mystery

A database breach occurred. The suspects are the **Frontend Service**, the **Legacy API**, the **Third-Party Plugin**, and the **DevOps Gateway**.

- The entity that caused the breach must have "Export" permissions.
- Only the **Legacy API**, the **Third-Party Plugin**, and the **DevOps Gateway** have "Export" permissions.
- An entity is "vulnerable" if it uses an outdated SSL version.
- The **Frontend Service** and the **DevOps Gateway** are not vulnerable.
- If an entity has Export permissions and is vulnerable, it is a "Primary Suspect."
- The **Third-Party Plugin** is vulnerable only if the **Frontend Service** is vulnerable.
- The **Legacy API** is always vulnerable.
- **Query:** Which entity is the Primary Suspect?

Scenario 3: The Deadlock Dilemma

A system-wide deadlock occurred, tracing back to one of six active entities: the **Main_Thread**, the **Worker_Pool**, the **Logger_Process**, the **UI_Handler**, the **Cleanup_Service**, or the **Network_Daemon**.

- The deadlock was triggered by an entity that currently holds a "Circular Wait" condition.
- According to the kernel scheduler, any entity holding a "Circular Wait" must also hold an active "Mutex Lock."
- Diagnostic logs show that the **Logger_Process** and the **Network_Daemon** do not hold Mutex Locks.
- The system's dependency graph indicates that the **Main_Thread** holds a Mutex Lock only if the **Worker_Pool** also holds one.
- Real-time monitoring confirms that the **Worker_Pool** and the **UI_Handler** are currently not holding any Mutex Locks.
- Due to an atomic operation, the **Cleanup_Service** holds a Mutex Lock if and only if the **UI_Handler** holds one.
- Every entity in the system, with the sole exception of the **Network_Daemon**, is currently flagged with a "Resource Request."
- The crash report guarantees that at least one of these six entities holds a Circular Wait.
- **Query:** Which specific entity caused the deadlock?

Scenario 4: The Zero-Day Exploit

A network is compromised. Use quantifiers to determine the state of the **Security Gateway**.

- For every user in the system, if that user accesses a malicious link, their workstation becomes infected.
- There exists at least one user who accessed a malicious link.
- Every infected workstation sends packets to the **Security Gateway**.
- If a packet is sent to the **Security Gateway** from an infected source, the gateway's firewall status becomes "compromised" unless the gateway is running the latest patch.
- No entity in the system is running the latest patch.
- **Query:** Prove the status of the Security Gateway.

Scenario 5: Distributed System Inconsistency

A distributed database is showing inconsistent data across its nodes.

- For every node in the cluster, if the node receives a "Write" request and does not send an "Acknowledgment" to the **Leader Node**, it is considered "out-of-sync."
- There exists a node that received a "Write" request but did not send an "Acknowledgment."
- For any two nodes, if one is "out-of-sync" and the other is a "Replica," the "Replica" will eventually time out.
- The **Backup Node** is a "Replica."
- The **Leader Node** is not "out-of-sync."
- **Query:** Prove that the Backup Node will eventually time out.

Scenario 6: The API Rate-Limit Failure

A service-level agreement (SLA) was violated due to an API failure.

- For every request sent to the **API Gateway**, there is a unique **Auth Token** associated with it.
- If any request is processed without a valid **Auth Token**, the system logs a "Security Violation."
- There exists a request that was processed by the **API Gateway**.
- For all requests processed by the **API Gateway**, their **Auth Tokens** were expired.
- An expired **Auth Token** is not a valid **Auth Token**.
- If a "Security Violation" is logged, the **On-Call Engineer** is alerted.
- **Query:** Prove the On-Call Engineer was alerted.

Scenario 7: The Orchestration Health Failure

A cluster orchestration failure has occurred. Prove that the **Global_Scheduler** has initiated a system-wide "Rollback" based on cascading failure logic and security protocol violations.

- For every microservice in the cluster, if that service is "unhealthy" and is not "restarted," then every dependent service will enter a "waiting" state.
- A microservice is considered "critical" if it handles sensitive user data; for every critical microservice, if it is in a "waiting" state, the **Security_Controller** must flag the entire namespace as "vulnerable."
- There exists a microservice named **Auth_Service** that is both "unhealthy" and "critical."
- **Rule of Propagation:** For every pair of microservices (A, B), if A is "unhealthy" and B is "dependent" on A, then B is "unhealthy" unless B has an active "Fallback_Circuit."
- No microservice in the cluster is currently "restarted," and no microservice has an active "Fallback_Circuit."
- For any microservice, if its namespace is flagged as "vulnerable," the **Monitoring_Agent** generates a "High_Priority_SLA_Violation" log for that service.
- If there exists at least one service for which a "High_Priority_SLA_Violation" log is generated, the **Global_Scheduler** initiates a "Rollback."
- **Query:** Prove that the **Global_Scheduler** initiated a "Rollback."

Question 3:

1. **Handwritten Solution:** For each requirement (A, B, and C), define your variables and derive the complete system of formal mathematical inequalities and equations from the descriptive text.

Output: If the system is **satisfiable**, provide the specific values for all four resource types. If the system is **unsatisfiable**, identify which specific constraint or combination of constraints causes the logical conflict.

2. **Code Submission & Screenshots:**

- Provide the SMT-LIB code (Z3) for each independent system using the variables you defined.
- Submit screenshots of the SMT-solver output.

A lead DevOps engineer is designing the resource footprint for three distinct microservice tiers. The design must account for four primary resource types: **Compute Units (vCPUs)**, **Memory Units (GB)**, **Storage Units (TB)**, and **Network Bandwidth (Gbps)**.

Note: Students must independently define their own variable names and manually derive all mathematical inequalities and equations from the descriptive requirements provided below. Each System Requirement (A, B, and C) is a standalone problem and must be solved as an independent SMT-LIB query.

System Requirement A: High-Availability Micro-Cluster

The provisioning for the High-Availability tier must satisfy the following architectural constraints:

- The total combined count of primary physical resources—Compute Units, Memory Units, and Storage Units—must be exactly 30 units.
- To maintain an optimal processing-to-RAM ratio, for every 2 units of Memory provided, the system must have at least 3 units of Compute allocated.
- The Storage allocation is strictly defined as exactly twice the count of Memory units minus a constant overhead of 5 units.
- For throughput stability, the allocated Network Bandwidth must be strictly greater than the sum of the Compute and Memory units.
- The thermal output of the rack, calculated by taking four times the Compute count, adding twice the Memory count, and adding the Storage count, must be strictly less than 85 units.
- **Safety Rule:** To prevent kernel panic during high load, the Memory units cannot be less than 6.

System Requirement B: Cost-Optimized Batch Processor

The provisioning for the Batch Processing tier must satisfy the following financial and operational constraints:

- **Total Cost Constraint:** The sum of triple the Compute units, four times the Memory units, and twice the Storage units must not exceed a total budget of 60.
- **Bandwidth Scaling:** The Network Bandwidth is dynamically calculated as the square of the Compute units plus an offset of 10.
- **Conditional Logic:** If the Compute units exceed a count of 4, then the Memory units must be strictly less than 4, and the Storage units must be at least 10.
- **Resource Parity:** To ensure balanced I/O, the difference between twice the Memory and the Storage units must be exactly 4 units.
- **Minimum Capacity:** The combined count of Compute and Storage units must be at least 15 units.

System Requirement C: Low-Latency Edge Node (Real/Float Logic)

The provisioning for the Edge Computing tier must satisfy the following latency-sensitive constraints using real-number logic:

- **Processing Latency:** Three-and-a-half times the Compute plus one-and-one-fifth times the Memory plus half the Storage must equal exactly 25.5 milliseconds.
- **Network Jitter Offset:** The Bandwidth must be tuned such that 80% of its value added to the Compute count equals exactly 15.0.
- **Ratio Calibration:** Half the Compute minus two-and-one-tenth times the Memory plus three-and-four-tenths times the Storage results in a specific calibration offset value of 12.2.
- **Hardware Requirement:** All resource units (Compute, Memory, Storage, and Bandwidth) must be strictly positive values.

Question 4:

Provide SMT-LIB code for the following HMS Scenario.

Consider the following “Knowledge base” of the hospital management system (HMS).

A) Handwritten:

- Identify Constants.
- Identify Variables.
- Identify Predicates.
- Identify Functions.

B) Code: Identify and declare relevant datatypes and their domains using the constants and variables previously identified ones. Declare the functions and predicates, and their input and output parameters using previously identified ones. Translate the following facts into assertions and then provide queries for questions at the end as well as screenshots of their answer from SMT-solver.

(Note: The code does not need to be handwritten for this question.)

HMS:

A hospital management system tracks patients, doctors, treatments, diseases, and prescriptions. The system stores data about which doctors treat which patients, the diseases patients have, and the medicines prescribed to treat those diseases. The goal is to manage the treatment process, track patient recovery, and ensure that appropriate doctors are assigned to the correct cases.

Doctor Information

1. **Dr. Smith** is a **Cardiologist**.
2. **Dr. Jones** is a **Neurologist**.
3. **Dr. Clark** is a **General Physician**.
4. **Dr. Evans** is a **Surgeon**.

Patient Information

1. **John Doe** is a **45-year-old male**.
2. **Mary Jane** is a **32-year-old female**.
3. **Jane Smith** is a **60-year-old female**.
4. **Michael Brown** is a **70-year-old male**.

Disease Information

1. **Cardiac Issues** are related to the **heart**.
2. **Migraine** is related to the **brain**.
3. **Flu** is classified under **general diseases**.
4. **Appendicitis** is also classified under **general diseases**.

Doctor's Specialization in Treating Diseases

1. **Dr. Smith** specializes in treating **Cardiac Issues**.
2. **Dr. Jones** specializes in treating **Migraine**.
3. **Dr. Clark** specializes in treating the **Flu**.
4. **Dr. Evans** specializes in treating **Appendicitis**.

Treatments and Medicines

1. **Cardiac Issues** can be treated with:
 - **Medication: Beta Blockers**
 - **Surgery: Angioplasty**
2. **Migraine** can be treated with:
 - **Medication: Pain Relief**
3. **Flu** can be treated with:
 - **Medication: Antiviral**
4. **Appendicitis** can be treated with:
 - **Surgery: Appendectomy**

Doctor-Patient Assignments

1. **Dr. Smith** is treating **John Doe** for **Cardiac Issues**.
2. **Dr. Jones** is treating **Mary Jane** for **Migraine**.
3. **Dr. Clark** is treating **Jane Smith** for the **Flu**.
4. **Dr. Evans** is treating **Michael Brown** for **Appendicitis**.

Prescriptions for Patients

Prescriptions are determined based on the specific **disease** each patient is diagnosed with. Below are the prescriptions aligned with their respective diseases:

1. **For Cardiac Issues:**
 - **Patient: John Doe**
 - **Doctor: Dr. Smith**
 - **Prescribed Medication: Beta Blockers**
 - **Prescribed Surgery (if needed): Angioplasty**
2. **For Migraine:**
 - **Patient: Mary Jane**
 - **Doctor: Dr. Jones**
 - **Prescribed Medication: Pain Relief**
3. **For Flu:**
 - **Patient: Jane Smith**
 - **Doctor: Dr. Clark**
 - **Prescribed Medication: Antiviral**
4. **For Appendicitis:**
 - **Patient: Michael Brown**
 - **Doctor: Dr. Evans**
 - **Prescribed Surgery: Appendectomy**

Hospital Rules for Critical Patients

1. **John Doe** is designated as a **critical patient**.
2. **Michael Brown** is also designated as a **critical patient**.

Provide Queries in SMT-LIB Syntax against each question.

1. Who is treating John Doe?
2. Which doctor specializes in treating appendicitis?
3. What medicines are prescribed for cardiac issues?
4. Is there any critical patient in the hospital?
5. What treatment is prescribed for migraine?
6. Which patients are being treated for general diseases?
7. What is the age and gender of John Doe?
8. Which doctor is prescribing beta blockers?
9. Is Dr. Jones treating any critical patients?
10. Who prescribes surgery for appendicitis?

Best of Luck ☺